NTC FAST RESPONSE THERMAL SENSORS

Sl. No	Product
1	Chip thermistor
2	Chip in glass fast response thermal sensors

ABOUT THE PRODUCTS:

NTC Chip thermistors are extensively used for accurate temperature measurement and control in automobiles, medical field and electronic appliances. To meet the challenging demands in the latest thermal devices high accuracy, close tolerance and high sensitivity thermistors are required

Nickel Manganite based spinel oxides are the most commonly used NTC materials. Doping with transition metal ions is a widely adopted technique to tune the electrical properties of the spinel oxides. By varying the types of dopants used, their relative proportions, the sintering atmosphere, and the sintering temperature; a wide range of resistivity and temperature coefficient characteristics can be obtained. The main characteristics of NTC thermistors are their specific resistivity, material constant (B-value) and the temperature coefficient of resistance (α).

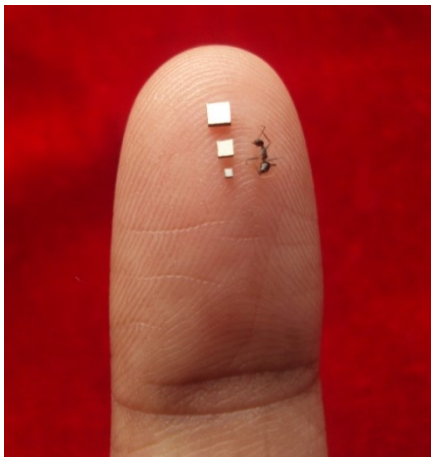
Now widely used thermistors are **epoxy coated chip thermistors** and **glass sealed bead thermistors**. Glass sealed bead thermistors are used for applications requiring small size, fast response time, high working temperature (up to 450°C), reliability and stability, and/or self heated use. The glass coating provides a true hermetic seal, critical for device stability, especially at use temperatures above 150°C but they have more unit-to-unit variation in resistance and Beta value than chip type thermistors. Individual calibration is usually required.

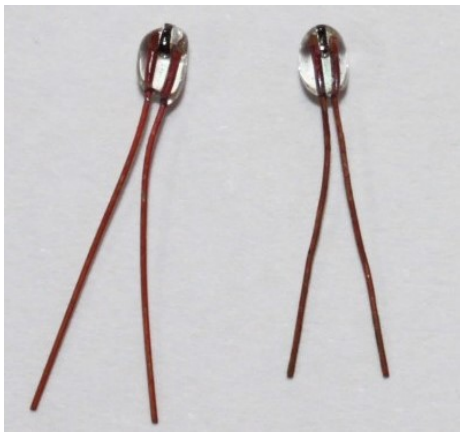
Epoxy coated NTC chip thermistors, with tight ‘resistance tolerance’ and ‘Beta characteristics’ can be used interchangeably for some applications eliminating the need for calibration but it do

not have the hermetic glass seal, thus limiting their stability and working temperatures to 150 ^{0}C or below.

Fast response Chip in glass NTC thermistor, an **alternative to glass sealed bead and epoxy coated chip NTC thermistors** combine the best characteristics of both devices. The advantages are

- Small size: 0.032" (0.8 mm) diameter
- Wide range of standard resistances
- Fast response time
- Excellent stability up to 250 ^{0}C
- Intermittent operation up to 450 ^{0}C
- Beta tolerances less than 1%
- Resistance tolerances 1 % to 15 %
- Available for high volume applications
- Lower cost
- Excellent stability of resistance
- High Precision Temperature Measurement





Specifications of Chip thermistors

Sl. No	Type	Material constant $B_{25/85}$ Value (K)	Resistivity (Ohm cm)	Size
1	Type I	3250 ± 0.75 %	2000± 7.5 %	0.5 mm x 0.5 mm x 0.5 mm (LxWxH) 1 mm x 1 mm x 0.5 mm (LxWxH) 1.5 mm x 1.5 mm x 0.5 mm (LxWxH)
2	Type II	3690 ± 0.75 %	2400± 7.5 %	0.5 mm x 0.5 mm x 0.5 mm (LxWxH) 1 mm x 1 mm x 0.5 mm (LxWxH) 1.5 mm x 1.5 mm x 0.5 mm (LxWxH)
3	Type III	3850 ± 0.75 %	2500± 7.5 %	0.5 mm x 0.5 mm x 0.5 mm (LxWxH) 1 mm x 1 mm x 0.5 mm (LxWxH) 1.5 mm x 1.5 mm x 0.5 mm (LxWxH)
4	Type IV	3977 ± 0.75 %	3100± 7.5 %	0.5 mm x 0.5 mm x 0.5 mm (LxWxH) 1 mm x 1 mm x 0.5 mm (LxWxH) 1.5 mm x 1.5 mm x 0.5 mm (LxWxH)
5	Type V	4252 ± 0.75 %	22000± 7.5 %	0.5 mm x 0.5 mm x 0.5 mm (LxWxH) 1 mm x 1 mm x 0.5 mm (LxWxH) 1.5 mm x 1.5 mm x 0.5 mm (LxWxH)

Technology Transfer

C-MET, Thrissur would like to invite competent industrial partners for commercialization of the above listed product through transfer of technology. The technology will be transferred on non-exclusive basis. The technology fee will be finalized at a later stage. The ToT package contains the following

1. Document for design and fabrication
2. Processing details
3. Product specification
4. Test plan and procedure
5. Training

C-MET, Thrissur will give training to the engineers of the ToT partner during different stages of the transfer of technology.

Who can apply?

Industries with good experience in manufacturing or assembling of thermal sensors can approach C-MET, Thrissur for the technology. Professionally managed Companies and Corporates are also welcome to apply for the technology. A Committee of experts constituted by the Department of Electronics and Information Technology (DeitY), Govt. of India will assess capabilities and strengths of the industry before finalizing the technology partners.

ToT Cost:

(i) The ToT cost for graphene supercapacitors is Rs10.07 lakhs plus applicable taxes.